```
!pip install spacy
!pip install transformers
!pip install torch
!pip install pandas
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spa
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spa
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=
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Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.4
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
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Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
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Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=
```

```
import spacy                    # NLP library for tokenization, POS tagging, and NER
from spacy import displacy      # Used to visualize named entities
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification  # Transformer-based NER
import torch                    # Deep learning backend for transformers
import pandas as pd             # Used to create tables (DataFrames)
```

```
nlp = spacy.load("en_core_web_sm")
```

```
sentences = [
    "Prime Minister Narendra Modi met President Joe Biden in Washington.",
    "Microsoft announced a new AI feature in Azure at the conference in California.",
    "Virat Kohli scored a century against Australia in Mumbai.",
    "The United Nations held a climate summit in New York.",
```

```
    "Tesla CEO Elon Musk launched Starship from Texas."
]
```

```
print("----- spaCy NER Results -----\n")

for sentence in sentences:
    doc = nlp(sentence)
    print("Sentence:", sentence)
    for ent in doc.ents:
        print("Entity:", ent.text, "| Label:", ent.label_)
    print("-" * 50)
```

```
----- spaCy NER Results -----

Sentence: Prime Minister Narendra Modi met President Joe Biden in Washington.
Entity: Narendra Modi | Label: PERSON
Entity: Joe Biden | Label: PERSON
Entity: Washington | Label: GPE
--------------------------------------------------
Sentence: Microsoft announced a new AI feature in Azure at the conference in California.
Entity: Microsoft | Label: ORG
Entity: AI | Label: ORG
Entity: California | Label: GPE
--------------------------------------------------
Sentence: Virat Kohli scored a century against Australia in Mumbai.
Entity: Virat Kohli | Label: PERSON
Entity: a century | Label: DATE
Entity: Australia | Label: GPE
Entity: Mumbai | Label: GPE
--------------------------------------------------
Sentence: The United Nations held a climate summit in New York.
Entity: The United Nations | Label: ORG
Entity: New York | Label: GPE
--------------------------------------------------
Sentence: Tesla CEO Elon Musk launched Starship from Texas.
Entity: Tesla | Label: ORG
Entity: Elon Musk | Label: PERSON
Entity: Texas | Label: GPE
--------------------------------------------------
```

```
data_spacy = []

for sentence in sentences:
    doc = nlp(sentence)
    for ent in doc.ents:
        data_spacy.append([sentence, ent.text, ent.label_])

df_spacy = pd.DataFrame(data_spacy, columns=["Sentence", "Entity", "Label"])
df_spacy
```

	Sentence	Entity	Label
0	Prime Minister Narendra Modi met President Joe...	Narendra Modi	PERSON
1	Prime Minister Narendra Modi met President Joe...	Joe Biden	PERSON
2	Prime Minister Narendra Modi met President Joe...	Washington	GPE
3	Microsoft announced a new AI feature in Azure ...	Microsoft	ORG
4	Microsoft announced a new AI feature in Azure ...	AI	ORG
5	Microsoft announced a new AI feature in Azure ...	California	GPE
6	Virat Kohli scored a century against Australia...	Virat Kohli	PERSON
7	Virat Kohli scored a century against Australia...	a century	DATE
8	Virat Kohli scored a century against Australia...	Australia	GPE
9	Virat Kohli scored a century against Australia...	Mumbai	GPE
10	The United Nations held a climate summit in Ne...	The United Nations	ORG
11	The United Nations held a climate summit in Ne...	New York	GPE
12	Tesla CEO Elon Musk launched Starship from Texas.	Tesla	ORG
13	Tesla CEO Elon Musk launched Starship from Texas.	Elon Musk	PERSON
14	Tesla CEO Elon Musk launched Starship from Texas.	Texas	GPE

```
doc = nlp(sentences[0])
displacy.render(doc, style="ent", jupyter=True)
```

Prime Minister Narendra Modi **PERSON** met President Joe Biden **PERSON** in Washington **GPE** .

```
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)

ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
config.json: 100%  998/998 [00:00<00:00, 16.0kB/s]
tokenizer_config.json: 100%  60.0/60.0 [00:00<00:00, 833B/s]
vocab.txt: 213k/? [00:00<00:00, 2.27MB/s]
model.safetensors: 100%  1.33G/1.33G [00:20<00:00, 196MB/s]
Loading weights: 100%  391/391 [00:02<00:00, 123.32it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                       | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.weight | UNEXPECTED |  |
bert.pooler.dense.bias   | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
print("----- Hugging Face NER Results -----\n")

for sentence in sentences:
    results = ner_pipeline(sentence)
    print("Sentence:", sentence)
    for res in results:
        print("Word:", res['word'],
              "| Label:", res['entity'],
              "| Score:", round(res['score'], 3))
    print("-" * 50)
```

```
----- Hugging Face NER Results -----

Sentence: Prime Minister Narendra Modi met President Joe Biden in Washington.
Word: Na | Label: I-PER | Score: 0.999
Word: ##ren | Label: I-PER | Score: 0.993
Word: ##dra | Label: I-PER | Score: 0.999
Word: Mo | Label: I-PER | Score: 0.999
Word: ##di | Label: I-PER | Score: 0.991
Word: Joe | Label: I-PER | Score: 0.999
Word: B | Label: I-PER | Score: 0.996
Word: ##iden | Label: I-PER | Score: 0.994
Word: Washington | Label: I-LOC | Score: 0.999
--------------------------------------------------
Sentence: Microsoft announced a new AI feature in Azure at the conference in California.
Word: Microsoft | Label: I-ORG | Score: 0.999
Word: A | Label: I-MISC | Score: 0.886
Word: ##zure | Label: I-ORG | Score: 0.575
Word: California | Label: I-LOC | Score: 0.999
--------------------------------------------------
Sentence: Virat Kohli scored a century against Australia in Mumbai.
Word: V | Label: I-PER | Score: 1.0
Word: ##ira | Label: I-PER | Score: 0.996
Word: ##t | Label: I-PER | Score: 0.999
Word: Ko | Label: I-PER | Score: 0.999
Word: ##hl | Label: I-PER | Score: 0.99
Word: ##i | Label: I-PER | Score: 0.994
Word: Australia | Label: I-LOC | Score: 1.0
Word: Mumbai | Label: I-LOC | Score: 0.999
--------------------------------------------------
```

```
Sentence: The United Nations held a climate summit in New York.
Word: United | Label: I-ORG | Score: 0.999
Word: Nations | Label: I-ORG | Score: 0.999
Word: New | Label: I-LOC | Score: 0.999
Word: York | Label: I-LOC | Score: 0.999
--------------------------------------------------
Sentence: Tesla CEO Elon Musk launched Starship from Texas.
Word: Te | Label: I-ORG | Score: 0.995
Word: ##sla | Label: I-ORG | Score: 0.986
Word: El | Label: I-PER | Score: 1.0
Word: ##on | Label: I-PER | Score: 0.999
Word: Mu | Label: I-PER | Score: 0.999
Word: ##sk | Label: I-PER | Score: 0.991
Word: Stars | Label: I-ORG | Score: 0.984
Word: ##hip | Label: I-ORG | Score: 0.987
Word: Texas | Label: I-LOC | Score: 1.0
--------------------------------------------------
```

```
cleaned_data = []

for sentence in sentences:
    results = ner_pipeline(sentence)

    for res in results:
        word = res['word'].replace("##", "")  # Remove subword symbols
        label = res['entity']
        score = round(res['score'], 3)

        cleaned_data.append([sentence, word, label, score])

df_hf = pd.DataFrame(cleaned_data,
                     columns=["Sentence", "Entity", "Label", "Confidence Score"])

df_hf
```

	Sentence	Entity	Label	Confidence Score
0	Prime Minister Narendra Modi met President Joe...	Na	I-PER	0.999
1	Prime Minister Narendra Modi met President Joe...	ren	I-PER	0.993
2	Prime Minister Narendra Modi met President Joe...	dra	I-PER	0.999
3	Prime Minister Narendra Modi met President Joe...	Mo	I-PER	0.999
4	Prime Minister Narendra Modi met President Joe...	di	I-PER	0.991
5	Prime Minister Narendra Modi met President Joe...	Joe	I-PER	0.999
6	Prime Minister Narendra Modi met President Joe...	B	I-PER	0.996
7	Prime Minister Narendra Modi met President Joe...	iden	I-PER	0.994
8	Prime Minister Narendra Modi met President Joe...	Washington	I-LOC	0.999
9	Microsoft announced a new AI feature in Azure ...	Microsoft	I-ORG	0.999
10	Microsoft announced a new AI feature in Azure ...	A	I-MISC	0.886
11	Microsoft announced a new AI feature in Azure ...	zure	I-ORG	0.575
12	Microsoft announced a new AI feature in Azure ...	California	I-LOC	0.999
13	Virat Kohli scored a century against Australia...	V	I-PER	1.000
14	Virat Kohli scored a century against Australia...	ira	I-PER	0.996
15	Virat Kohli scored a century against Australia...	t	I-PER	0.999
16	Virat Kohli scored a century against Australia...	Ko	I-PER	0.999
17	Virat Kohli scored a century against Australia...	hl	I-PER	0.990
18	Virat Kohli scored a century against Australia...	i	I-PER	0.994
19	Virat Kohli scored a century against Australia...	Australia	I-LOC	1.000